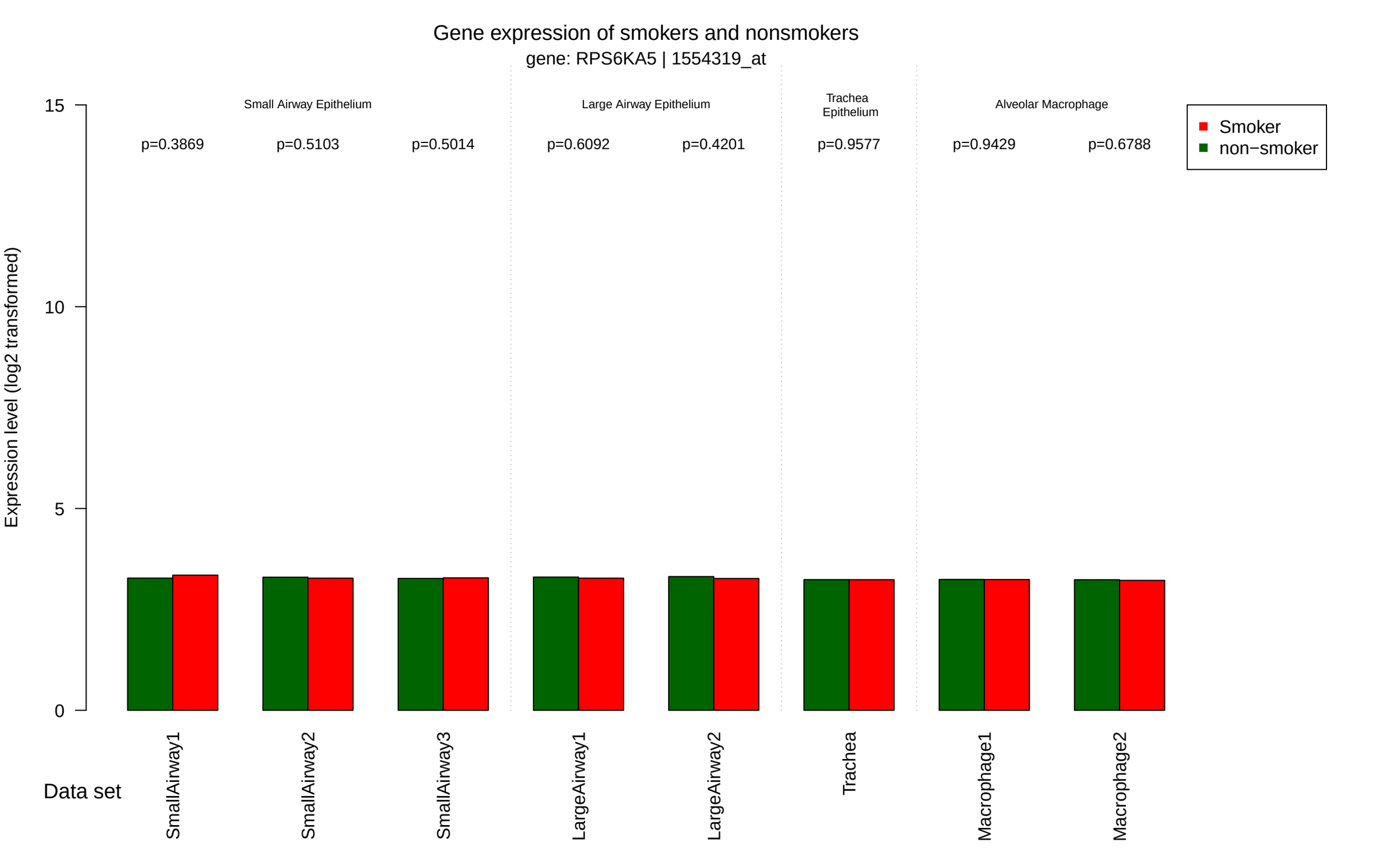
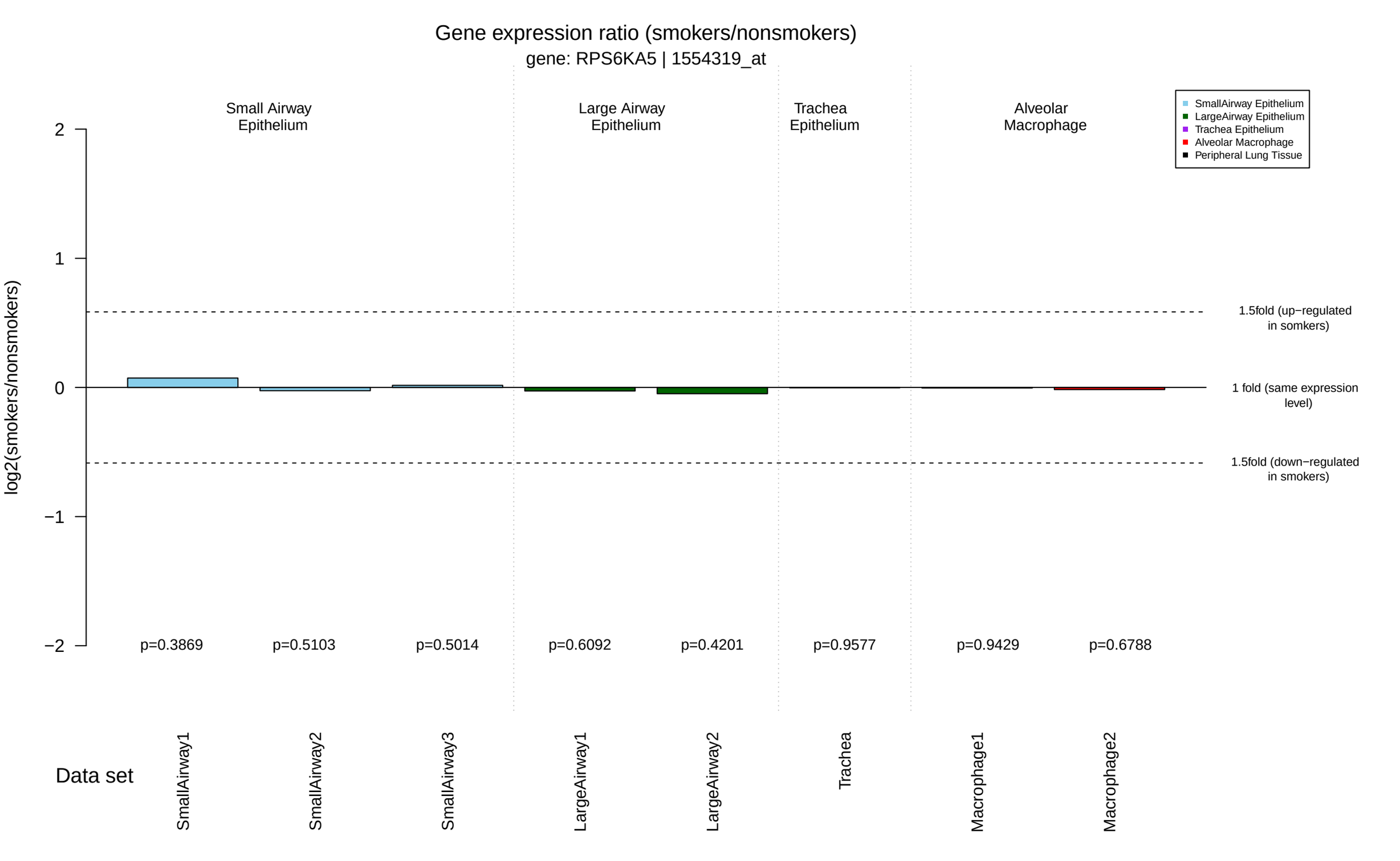
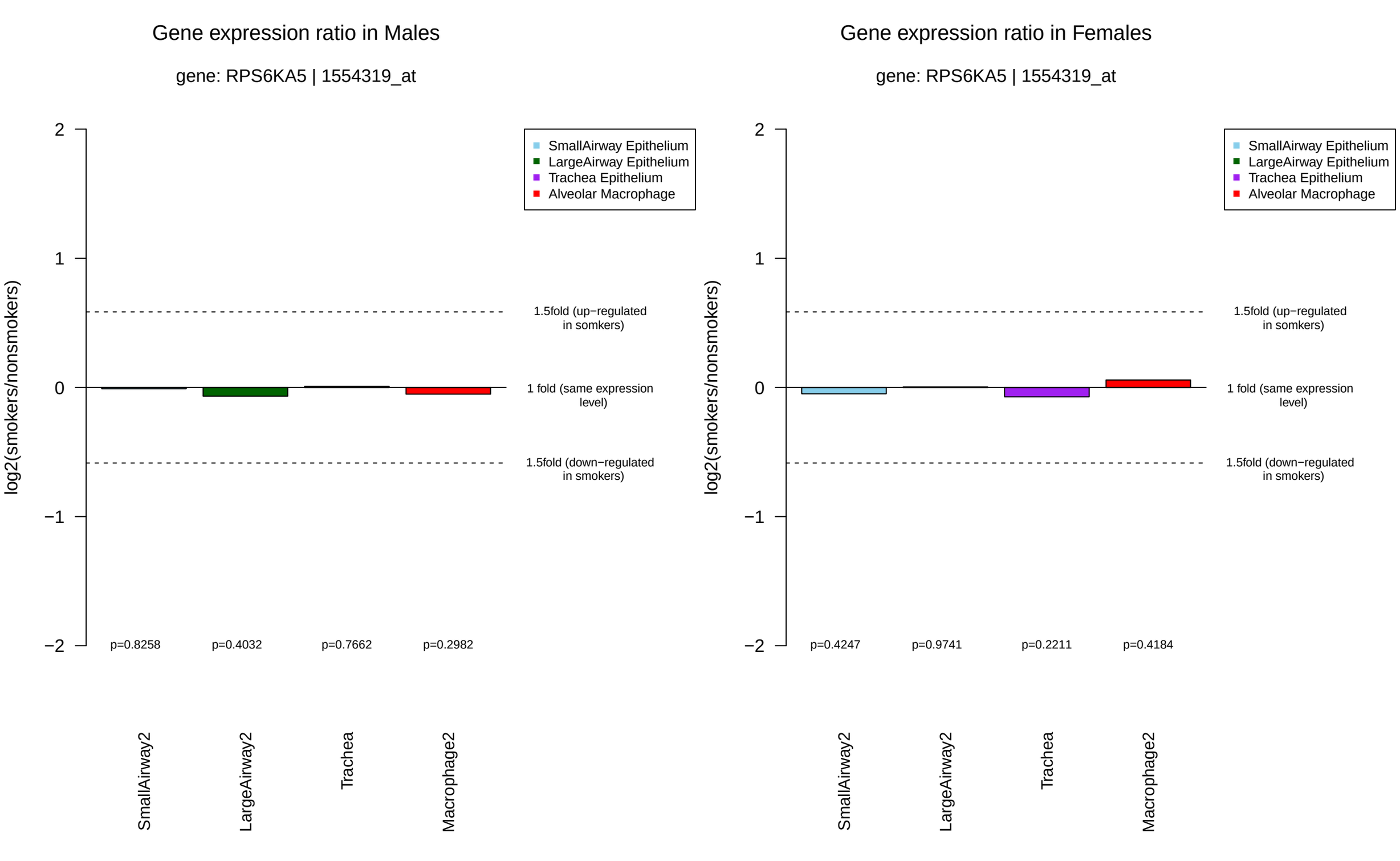
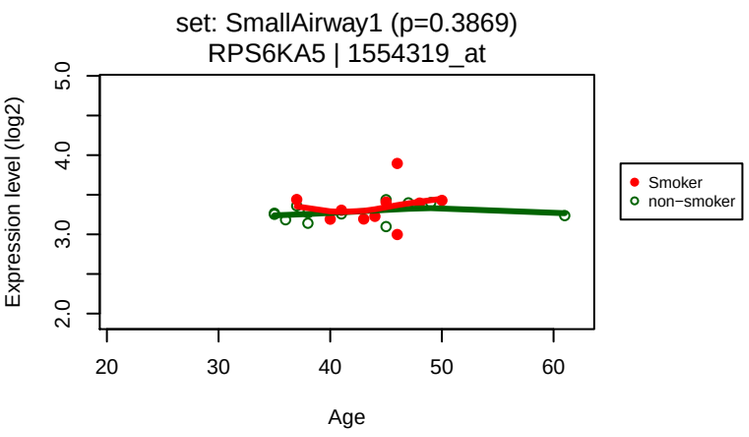
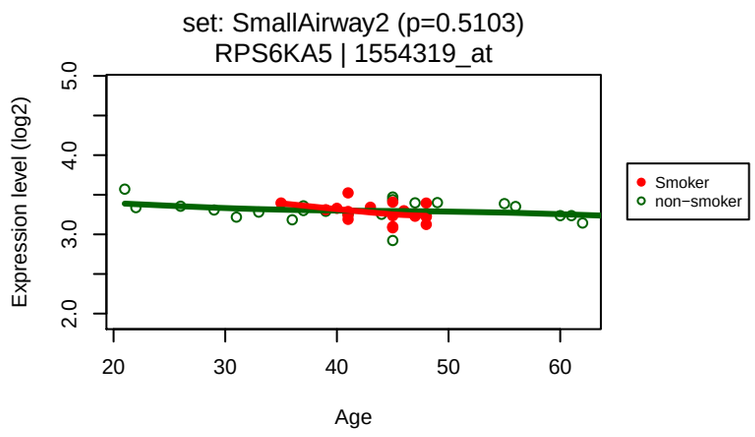
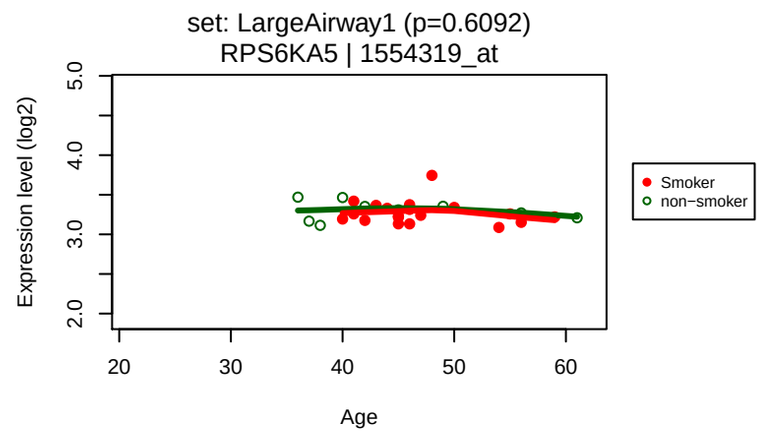

Gene expression of smokers and nonsmokers
gene: RPS6KA5 | 1554319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3869
p=0.5103
p=0.5014
p=0.6092
p=0.4201
p=0.9577
p=0.9429
p=0.6788
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

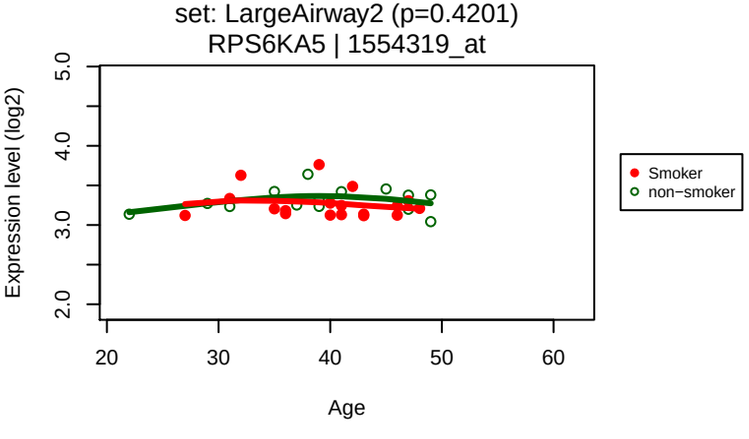

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA5 | 1554319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3869
p=0.5103
p=0.5014
p=0.6092
p=0.4201
p=0.9577
p=0.9429
p=0.6788
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

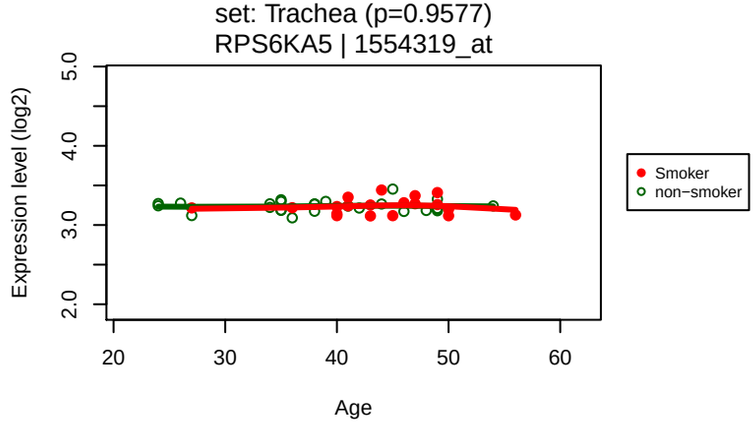

Gene expression ratio in Males
gene: RPS6KA5 | 1554319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8258
p=0.4032
p=0.7662
p=0.2982
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA5 | 1554319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4247
p=0.9741
p=0.2211
p=0.4184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

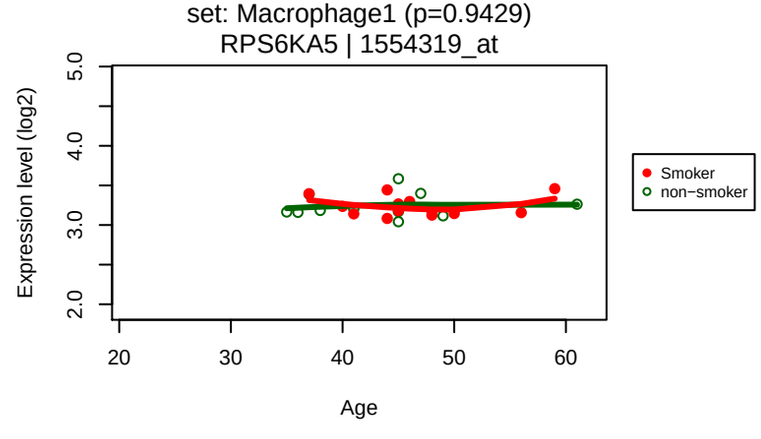

set: SmallAirway1 (p=0.3869)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker

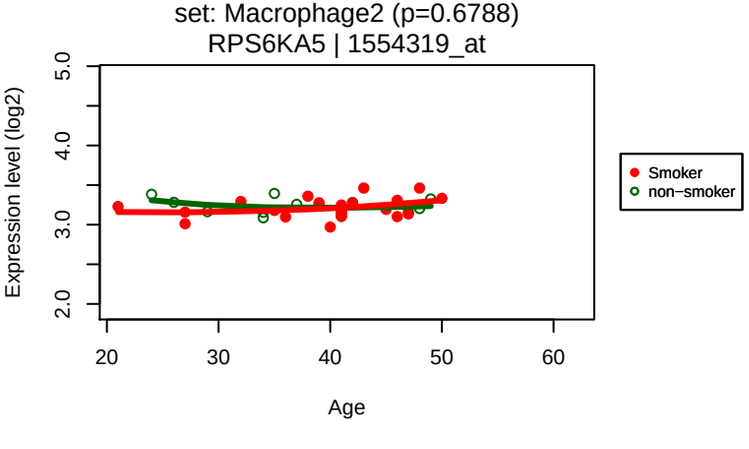

set: SmallAirway2 (p=0.5103)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker

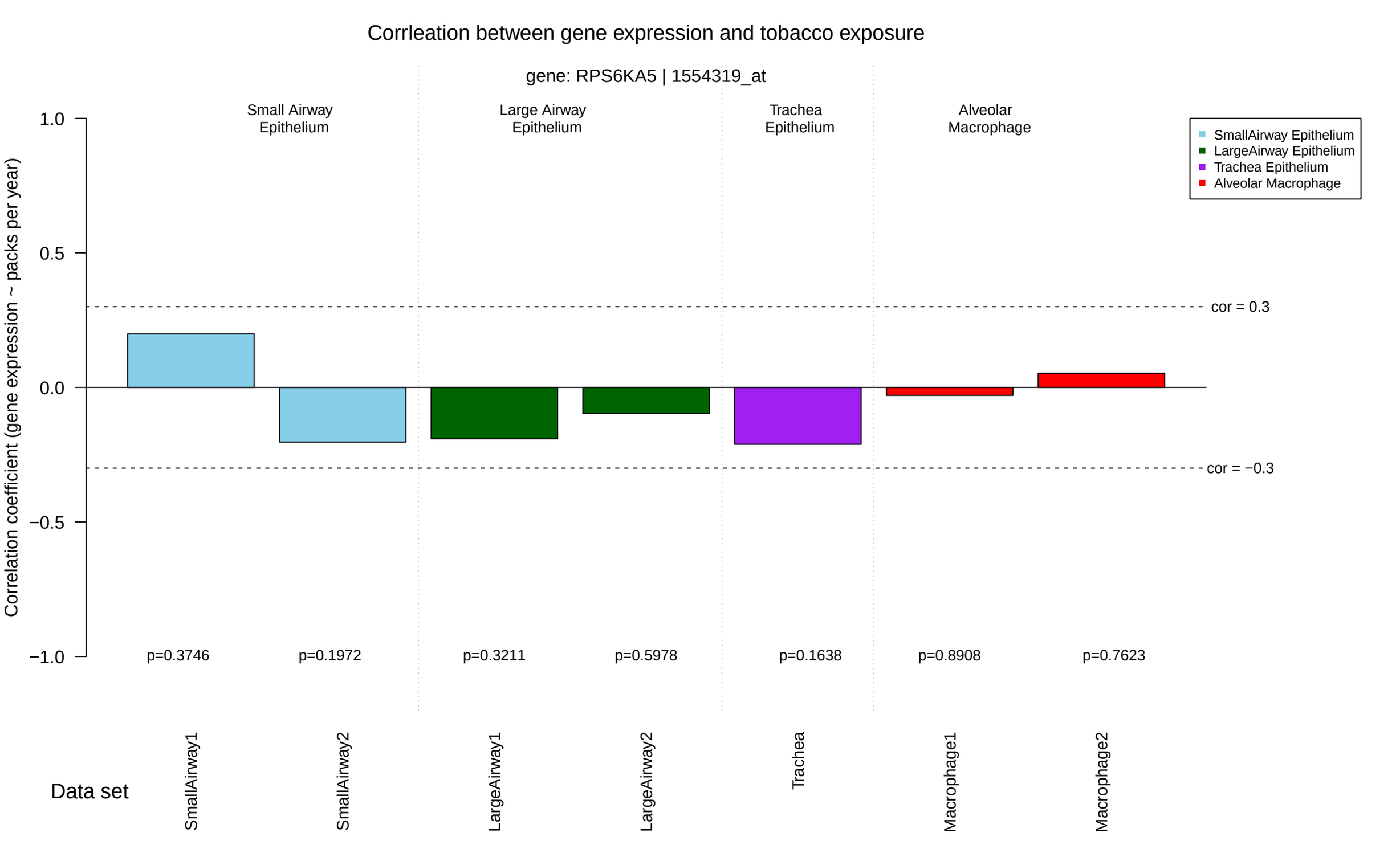

set: LargeAirway1 (p=0.6092)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4201)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9577)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9429)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6788)
RPS6KA5 | 1554319_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA5 | 1554319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3746
p=0.1972
p=0.3211
p=0.5978
p=0.1638
p=0.8908
p=0.7623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2